

VIRGINIA TECH MONTGOMERY, VA, USA

WMO: 724113

Lat: 37.208N

Lon: 80.408W

Elev: 2132

StdP: 13.60

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 53881

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 10.3	(c) 15.7	(d) -2.2	(e) 5.3	(f) 17.6	(g) 1.7	(h) 6.5	(i) 18.9	(j) 24.7	(k) 27.8	(l) 21.9	(m) 28.9	(n) 7.7	(o) 300	(p) 0.419

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.3	(c) 88.8	(d) 72.3	(e) 86.3	(f) 71.3	(g) 83.9	(h) 70.6	(i) 75.6	(j) 83.4	(k) 74.3	(l) 81.8	(m) 73.0	(n) 80.2	(o) 5.8	(p) 300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.2	133.5	78.6	72.3	129.4	77.8	71.1	124.1	76.8	40.7	82.7	39.3	81.9	38.2	80.4	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.7	(b) 17.2	(c) 15.0	(e) 4.8	(f) 91.9	(g) 4.8	(h) 2.7	(i) 1.3	(j) 93.8	(k) -1.5	(l) 95.3	(m) -4.2	(n) 96.8	(o) -7.7	(p) 98.8		
(4) 19.7	(5) 17.2	(6) 15.0	(7) 4.8	(8) 91.9	(9) 4.8	(10) 2.7	(11) 1.3	(12) 93.8	(13) -1.5	(14) 95.3	(15) -4.2	(16) 96.8	(17) -7.7	(18) 98.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.3	33.7	36.9	44.3	54.2	62.6	69.8	73.2	72.2	66.0	55.4	44.6	37.3
	DBStd	16.01	10.52	9.82	9.60	8.42	7.06	5.17	4.24	4.39	6.28	7.93	8.65	9.01
	HDD50	1828	514	378	229	52	4	0	0	0	39	206	406	
	HDD65	4730	969	787	642	332	129	18	2	4	63	312	613	859
	CDD50	3388	10	11	52	178	396	595	718	689	479	205	44	11
	CDD65	817	0	0	0	8	56	164	256	228	92	13	0	0
	CDH74	6375	0	1	14	162	503	1248	1951	1671	707	111	7	0
	CDH80	1762	0	0	0	24	89	329	620	505	179	16	0	0
	WSAvg	4.9	6.7	6.6	6.7	6.2	4.6	3.8	3.3	3.1	3.5	4.3	5.1	5.6
	PrecAvg	41.0	3.0	3.0	3.7	3.5	4.2	3.6	4.2	3.7	3.4	3.1	2.9	3.0
(15) Precipitation	PrecMax	57.6	7.4	6.3	8.3	7.8	9.8	8.7	9.4	7.5	8.9	9.1	7.4	6.5
	PrecMin	28.7	0.6	0.4	0.6	0.7	0.9	0.4	1.6	0.9	0.2	0.0	0.4	0.2
	PrecStd	6.8	1.3	1.5	1.6	1.6	2.0	2.0	1.7	1.4	2.0	1.9	1.6	1.5
	PrecStd	6.8	1.3	1.5	1.6	1.6	2.0	2.0	1.7	1.4	2.0	1.9	1.6	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	64.9	68.2	75.4	82.3	85.7	90.3	91.2	91.0	88.4	81.7	73.3	66.0	
	0.4% MCWB	56.9	57.8	59.6	63.5	67.5	73.0	74.2	73.0	68.1	66.3	59.1	58.6	
	2% DB	60.7	62.9	70.9	79.0	82.1	86.4	88.5	88.3	84.3	77.1	67.9	61.2	
	2% MCWB	54.4	52.4	57.0	61.3	67.0	71.2	73.2	72.4	68.8	64.3	56.3	54.9	
	5% DB	55.0	57.7	65.4	74.6	80.2	83.7	86.1	85.4	81.6	72.8	63.7	56.7	
	5% MCWB	49.4	49.4	53.9	59.3	66.1	70.8	72.1	71.7	68.0	61.8	55.5	51.7	
	10% DB	50.3	53.7	61.2	70.5	76.5	81.4	83.6	82.2	78.7	69.7	60.1	52.4	
	10% MCWB	45.7	46.7	51.7	57.5	64.6	69.5	71.4	70.4	67.2	61.3	53.0	47.8	
	0.4% WB	59.9	60.3	62.3	66.7	72.2	76.6	77.9	77.6	74.5	71.1	63.3	61.0	
	0.4% MCDB	63.8	65.3	71.8	77.6	79.7	85.0	86.4	83.6	81.0	76.5	69.6	63.8	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	55.0	55.5	59.3	64.2	70.1	73.8	76.1	75.3	72.4	67.8	60.0	57.0	
	2% MCDB	59.2	60.6	66.9	73.8	78.2	81.8	84.2	82.8	78.7	73.1	64.9	59.8	
	5% WB	50.8	51.3	56.5	61.9	68.2	72.3	74.5	73.9	70.4	64.7	57.1	53.2	
	5% MCDB	54.6	56.2	63.2	70.9	76.4	80.4	82.2	81.1	76.8	70.2	61.6	55.8	
	10% WB	45.5	46.7	53.5	59.5	66.4	70.8	73.1	72.5	68.8	62.1	53.9	47.9	
	10% MCDB	49.3	52.0	59.1	68.0	73.8	78.8	80.2	79.5	75.4	67.5	58.7	51.1	
(35) Mean Daily Temperature Range	MDBR	17.6	18.7	20.9	22.8	21.4	19.9	19.3	19.5	20.3	21.5	21.2	17.7	
	5% DB MCDBR	24.1	27.2	28.9	29.9	26.0	23.7	23.0	23.4	25.2	27.0	27.9	24.6	
	5% DB MCWBR	17.7	18.8	16.8	15.1	12.6	10.7	9.6	9.9	11.6	14.7	17.9	18.0	
	5% WB MCDBR	21.1	23.4	23.9	25.0	21.9	20.3	19.3	19.5	20.2	20.7	21.8	21.2	
	5% WB MCWBR	17.5	18.5	15.8	14.0	11.5	10.0	9.2	9.2	10.2	12.8	16.2	17.8	
	5% WB MCWBR	17.5	18.5	15.8	14.0	11.5	10.0	9.2	9.2	10.2	12.8	16.2	17.8	
(40) Clear-Sky Solar Irradiance	taub	0.293	0.307	0.332	0.375	0.413	0.445	0.475	0.477	0.378	0.342	0.309	0.293	
	taud	2.509	2.459	2.435	2.335	2.233	2.176	2.122	2.091	2.371	2.464	2.524	2.539	
	Ebn at Noon	287	295	296	287	277	266	257	254	277	278	278	279	
	Edh at Noon	25	30	34	39	44	47	49	50	36	30	24	23	
	Edh at Noon	25	30	34	39	44	47	49	50	36	30	24	23	
(44) All-Sky Solar Radiation	RadAvg	709	932	1265	1605	1769	1928	1810	1656	1389	1102	822	615	
	RadStd	60	112	81	129	151	139	113	85	136	135	78	57	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.86	N/A	N/A	N/A	+1.04	+1.30	N/A	N/A	+265	N/A
(47) Regional (40 neighbors)		+0.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+238	+128

Nomenclature: See separate page